ROAD MAP

Title: Explainable Real Estate Analytics: A Comparative Analysis of Machine Learning Frameworks for House Price Prediction

Domain

Real Estate Analytics | Machine Learning | Data Science

Dataset

I recommend using benchmark house price datasets:

1. **House price prediction (Kaggle)**
   - https://www.kaggle.com/datasets/shree1992/housedata
2. **House Price Prediction Dataset (Kaggle)**
   - https://www.kaggle.com/datasets/zafarali27/house-price-prediction-dataset
3. **House Prices India (Kaggle)**
   - https://www.kaggle.com/datasets/sukhmandeepsinghbrar/house-prices-india?utm_source=chatgpt.com

These datasets contain important housing attributes such as area, number of bedrooms, bathrooms, location, year built, lot size, and selling price, making them suitable for comparative machine learning analysis.

Abstract

The real estate industry generates large volumes of property-related data that can be leveraged to estimate house prices accurately. Traditional valuation methods often rely on expert knowledge and manual analysis, which may produce inconsistent results. This project proposes a machine learning-based framework for House Price Prediction by performing comprehensive Exploratory Data Analysis (EDA)**,** data preprocessing, feature engineering, and comparative evaluation of multiple machine learning models.

Initially, the collected housing datasets undergo preprocessing, including handling missing values, removing duplicate records, encoding categorical attributes, and normalizing numerical features. A detailed Exploratory Data Analysis (EDA) is then conducted to understand the data through dataset overview, descriptive statistics, data quality assessment, missing value analysis, duplicate detection, outlier detection, univariate analysis, bivariate analysis, multivariate analysis, correlation analysis, feature engineering, and feature selection.

Multiple machine learning algorithms, including Decision Tree, Random Forest, XGBoost, and CatBoost, are trained and evaluated for predicting house prices. Their performance is compared using MAE, MSE, RMSE, R^2 Score, and MAPE, and the best-performing model is selected.

To improve model interpretability, the project incorporates SHAP and LIME**,** providing both global and local explanations of predictions. These techniques help identify the most influential features affecting house prices.

Finally, the analytical results are presented through interactive visualizations such as heatmaps, boxplots, feature importance graphs, correlation plots, and dashboards. A comparative analysis is conducted across all machine learning models to determine the most accurate and interpretable approach for house price prediction.

Approach:

Problem Definition

•Define project objectives

• Collect the house price dataset

• Understand dataset attributes and target variable

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Dataset Overview

• Shape

• Features

• Data Types

• Target Variable

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Data Quality Assessment

• Missing Values

• Duplicate Records

• Inconsistent Data

• Outlier Detection

• Encode categorical features

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Exploratory Data Analysis (EDA)

• Descriptive Statistics

• Univariate Analysis

• Bivariate Analysis

• Multivariate Analysis

• Correlation Analysis

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Feature Engineering

• Create New Features

• Transform Variables

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Feature Selection

• Correlation-Based Selection

• Feature Importance

• Remove Irrelevant Features

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Data Preprocessing

• Missing Value Imputation

• Encoding Categorical Features

• Feature Scaling

• Data Cleaning

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Train-Test Split

• Training Dataset (80%)

• Testing Dataset (20%)

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Model Training

- Decision Tree
- Random Forest
- XGBoost
- CatBoost

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Model Evaluation

• MAE

• MSE

• RMSE

• R^2 Score

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Comparative Analysis

• Compare Model Performance

• Select Best Model

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Model Explainability

• SHAP Analysis

• LIME Analysis

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Visualization

• Heatmap

• Boxplot

• Feature Importance

• Prediction vs Actual

• Dashboard

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Final House Price Prediction

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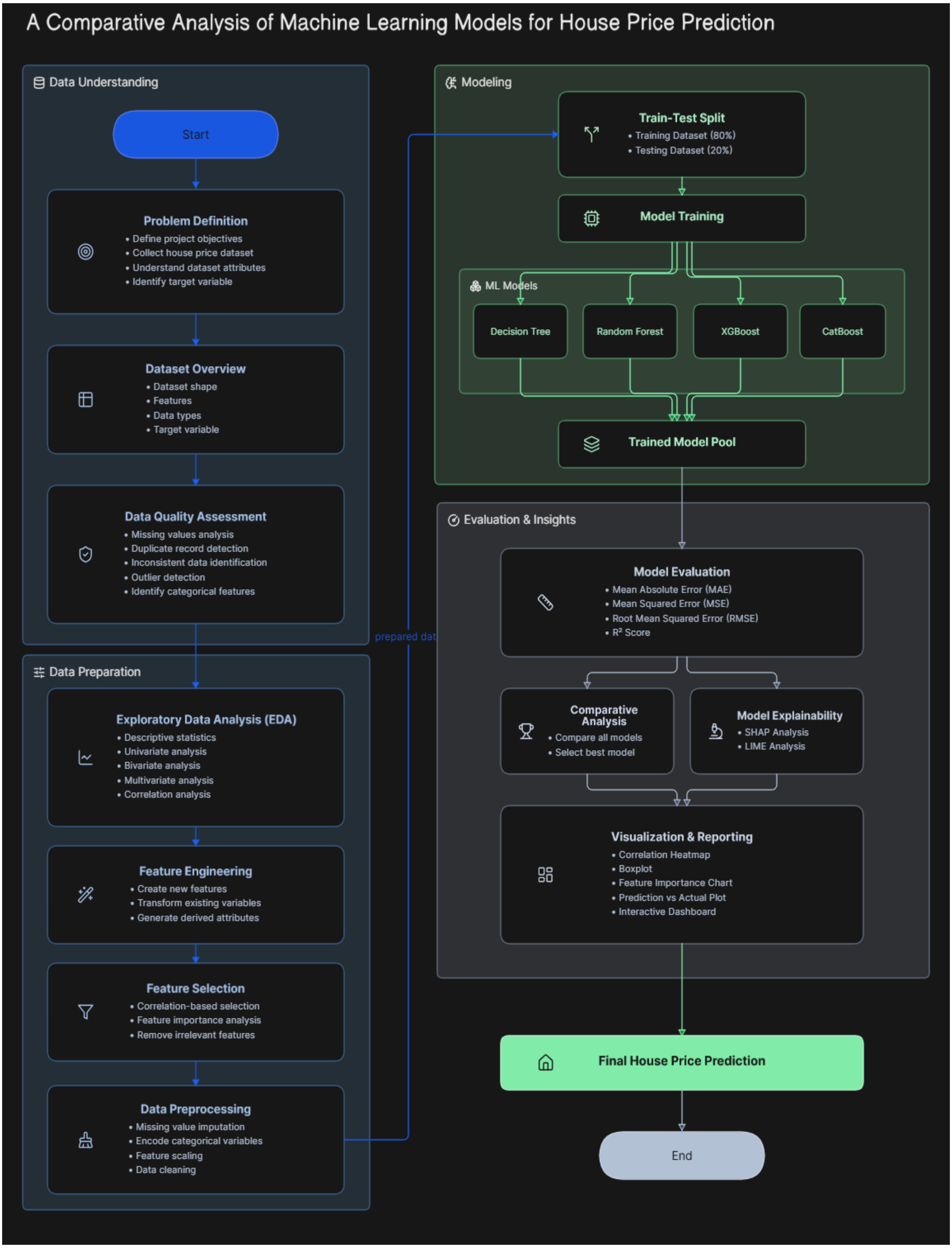
End


A Comparative Analysis of Machine Learning Models for House Price Prediction
Data Understanding
Start
Problem Definition
• Define project objectives
• Collect house price dataset
• Understand dataset attributes
• Identify target variable
Dataset Overview
• Dataset shape
• Features
• Data types
• Target variable
Data Quality Assessment
• Missing values analysis
• Duplicate record detection
• Inconsistent data identification
• Outlier detection
• Identify categorical features
Data Preparation
Exploratory Data Analysis (EDA)
• Descriptive statistics
• Univariate analysis
• Bivariate analysis
• Multivariate analysis
• Correlation analysis
Feature Engineering
• Create new features
• Transform existing variables
• Generate derived attributes
Feature Selection
• Correlation-based selection
• Feature importance analysis
• Remove irrelevant features
Data Preprocessing
• Missing value imputation
• Encode categorical variables
• Feature scaling
• Data cleaning
prepared dat
Modeling
Train-Test Split
• Training Dataset (80%)
• Testing Dataset (20%)
Model Training
ML Models
Decision Tree
Random Forest
XGBoost
CatBoost
Trained Model Pool
Evaluation & Insights
Model Evaluation
• Mean Absolute Error (MAE)
• Mean Squared Error (MSE)
• Root Mean Squared Error (RMSE)
• R² Score
Comparative Analysis
• Compare all models
• Select best model
Model Explainability
• SHAP Analysis
• LIME Analysis
Visualization & Reporting
• Correlation Heatmap
• Boxplot
• Feature Importance Chart
• Prediction vs Actual Plot
• Interactive Dashboard
Final House Price Prediction
End

Preprocessing

- Missing Value Handling
- Duplicate Removal
- Encoding
- Normalization / Standardization

Machine Learning Models

- Decision Tree Regressor
- Random Forest Regressor
- XGBoost Regressor
- CatBoost Regressor

Explainability

- SHAP
- LIME

Visualization

- Heatmap
- Boxplot
- Correlation Matrix
- Feature Importance
- Distribution Plots
- Dashboard

Comparative Analysis

Compare Models

- Decision Tree
- Random Forest
- XGBoost
- CatBoost

Evaluation Metrics

- MAE (Mean Absolute Error)
- MSE (Mean Squared Error)
- RMSE (Root Mean Squared Error)
- R^2 Score
- MAPE

Recommended Research Pipeline:

Benchmark Dataset

Data Cleaning

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Exploratory Data Analysis (EDA)

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Feature Engineering

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Feature Selection

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Train-Test Split

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Machine Learning Models
(Decision Tree
Random Forest
XGBoost
CatBoost)

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Performance Evaluation
(MAE, MSE, RMSE, R^2, MAPE)

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Explainable AI
(SHAP + LIME)

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Visualization Dashboard

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Comparative Analysis

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Research Conclusions

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